Fiscal Year:	FY 2011	**Task Last Updated:**	FY 05/04/2011
PI Name:	Wang, Ya M.D., Ph.D.		
Project Title:	The mechanism of excess relative risk on carcinogenesis induced by high-LET radiation		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
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Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
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No. of PhD Candidates:		**No. of Master' Degrees:**	
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Guida, Peter Ph.D. (BNL)		
Grant/Contract No.:	NNX09AF24G		
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Task Description:	Our hypothesis in this application is that high LET radiation induced excess relative risk on mutagenesis as well as on carcinogenesis is due to inefficient Ku-dependent non-homologous end-joining (NHEJ) repair. The results from this project will provide a mechanism to explain why high LET radiation induced such excess relative risk (ERR), and will significantly improve our understanding of the relationship between DNA repair and high LET radiation induced such ERR. Therefore, the results from this proposal will assist NASA Space Radiation Program to reach its major goal by providing specific parameters for evaluating high LET radiation induced ERR on mutagenesis as well as on carcinogenesis and by providing protective targets for reducing high LET radiation induced mutagenesis as well as carcinogenesis.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The study described in this application will determine whether the major reason for high linear energy transfer (LET) ionizing radiation (IR) to induce excess relative risk (ERR) as compared with low LET IR on carcinogenesis is because of inefficient Ku-dependent non-homologous end-joining repair. The results from this proposal will answer the question whether inefficient Ku-dependent repair is the major reason for high LET IR to induce ERR on carcinogenesis. Results from this proposal will assist the NASA Space Radiation Program to reach its major goal by providing specific parameters for evaluating the ERR on carcinogenesis and by providing specific protective targets for reducing high LET IR-induced ERR on the carcinogenesis.
Task Progress:	This project includes two major categories: cell experiments and animal experiments. The animal work includes survival (acute) and carcinogenesis (chronic) experiments. All our planned work is finished and the results are summarized as follows: 1. Cell work: We carried out the experiments by using different types of cells and obtained the following interesting results: 1) We elucidate why S-phase cells are more sensitive to heavy ion, the high linear energy transfer (LET) radiation (component of space radiation), although they are more resistant to x or gamma ray. After further studying the mechanism, we believe that the phenotype of S-phase sensitive to high-LET radiation is due to the fact that high-LET radiation could generate much smaller fragments (< 40 base pairs) that result in inefficient Ku binding in S –phase, which further supports our hypothesis that high-LET killing more cells when compared with low-LET radiation is because of the inhibition of Ku-dependent NHEJ. This work is published in Red Journal (Wang H, Liu S, Zhang P, Zhang S, Naidu M, Wang H, Wang Y. "S-phase cells are more sensitive to high-linear energy transfer radiation." Int J Radiat Oncol Biol Phys. 2009 Jul 15;74(4):1236-41.). 2) We demonstrate that the checkpoint response plays a more protective role in HZE particle-irradiated cells than in X ray-irradiated cells. We previously reported that the DNA double strand break (DSB)-activated checkpoint mainly facilitates homologous recombination repair (HRR) and has little effect on non-homologous end-joining (NHEJ). After combining our new findings with high-LET radiation, we tested our hypothesis that the checkpoint response plays a more protective role in HZE particle-irradiated cells than in X ray-irradiated cells and obtained supportive results. This work is published in Cell Cycle. 3) We indicate that low dose radiation-induced adaptive responses are through promoting NHEJ and explain that low dose radiation could not induce any protective role in high-LET irradiated cells. It has been known for more than two decades that low dose radiation (< 0.1 Gy) could induce adaptive responses. The adaptive responses could protect cells and animals from high-challenge dose-induced killing and carcinogenesis. Because DNA DSBs induced by radiation are the severest threat for cell survival and genetic integrity, the low dose adaptive responses are believed to promote DNA repair. However, it remains unclear which DNA DSB repair pathway, NHEJ or HRR, is promoted by the low dose radiation. We found that low dose radiation could not induce any protective role in high-LET irradiated cells. After combining these results with our repair theory about high-LET radiated cells, we hypothesized that low dose radiation-induced adaptive responses are through promoting NHEJ. We tested this hypothesis and obtained supportive results. This work is published in the Journal of Cell Physiology (Yu X, Wang H, Wang P, Chen BP, Wang Y. "The Ku dependent non-homologous end-joining pathway contributes to low-dose radiation-stimulated cell survival." J Cell Physiol. 2011 Feb;226(2):369-74.). 4) We show for the first time that miR-21 plays an important role in HZE particle-induced carcinogenesis. By obtaining the paraffin-embedded liver tissues from irradiated mice, which were provided by Dr. Robert L. Ullrich (University of Texas Medical Branch), Dr. Michael Weil (Colorado State University), Drs. Gayle Woloschak and Tanja Paunesku (Northwestern University), we found that microRNA-21 (miR-21) was much higher expressed in liver tumors than in normal liver tissues and further found that ionizing radiation, especially high LET radiation could stimulate miR-21 expression. We showed that after the non-irradiated, low-LET or high-LET irradiated human hepatocytes were over-expressed with miR-21, these cells became tumorigenesis in nude mice. The tumors derived from high-LET-irradiated-cells were largest, and accompanied by more significant changes in the miR-21-targets: PTEN and RECK. In addition, we showed that IR-induced up-regulation of miR-21 depended on the up-regulation/activation of AP-1 (at an earlier time, within 2 h) and the ErbB/Stat3 pathway (at a later time, more than 2 h), which was also IR dose dependent. Taken together, we conclude that IR-induced up-regulation of miR-21 plays an important role in IR (especially high-LET IR)-promoted liver carcinogenesis. This work is published in Int J Clin Exp Med (Zhu Y, Yu X, Fu H, Wang H, Wang P, Zheng X, Wang Y. "MicroRNA-21 is involved in ionizing radiation-promoted liver carcinogenesis." Int J Clin Exp Med. 2010 Jul 31;3(3):211-22.). 2. Animal work: We carried out the experiments by using mice with different genetic backgrounds and obtained the following interesting results: 1) Acute experiments: we demonstrate that more effects of high-LET radiation than low-LET radiation-induced killing is due to the inefficient NHEJ repair, which is involved in Mre11 and Ku responding differently to shorter DNA fragments in vitro and to the DNA from high-LET irradiated cells in vivo. These findings provide strong evidence that the different DNA DSB binding properties of Mre11 and Ku determine the different efficiencies of HRR and NHEJ to repair high-LET radiation-induced DSBs. This work is published in Nucleic Acid Res (Wang H, Zhang X, Wang P, Yu X, Essers J, Chen D, Kanaar R, Takeda S, Wang Y. "Characteristics of DNA-binding proteins determine the biological sensitivity to high-linear energy transfer radiation." Nucleic Acids Res. 2010 Jun;38(10):3245-51.). 2) Chronic experiments: we elucidate that DNA repair plays an important role in protecting mice from HZE particle-induced carcinogenesis. Our results showed that ionizing radiation increased tumorigenesis, especial lung tumorigenesis. These phenotypes were more clearly shown in DNA repair (HRR or NHEJ) deficient mice. However, we did not see dramatically different frequencies of carcinogenesis between high-LET and low-LET irradiated mice although the tumorigenesis frequency in DNA-PKcs-/- (NHEJ deficient) mice is higher than in Rad54-/- (HRR deficient) mice and the tumorigenesis frequency in Rad54-/- mice is higher than in wild type mice. These results indicate that both NHEJ and HRR protect mice from radiation-induced carcinogenesis but besides DNA repair, other factors also contribute to radiation-induced carcinogenesis. Although acute high-LET radiation-induced more killings then low-LET radiation, high-LET radiation does not show a dramatically higher carcinogenesis frequency than low-LET radiation. In addition, except high-LET fractionation radiation (0.2 Gy x5) that showed higher liver carcinogenesis than high-LET

	single dose (1 Gy) irradiated in DNA-PKcs-/- mice, fractionation radiation (0.2 Gy x5) did not show a dramatically different frequency for other types of mouse tumors when compared with single dose (1 Gy) irradiation.
Bibliography Type:	Description: (Last Updated: 07/07/2021)
Articles in Peer-reviewed Journals	Wang H, Wang X, Zhang P, Wang Y. "The Ku-dependent non-homologous end-joining but not other repair pathway is inhibited by high-linear energy transfer ionizing radiation." DNA Repair (Amst). 2008 May 3;7(5):725-33. Epub 2008 Mar 5. PMID: 18325854 , May-2008
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Articles in Peer-reviewed Journals	Zhu Y, Yu X, Fu H, Wang H, Wang P, Zheng X, Wang Y. "MicroRNA-21 is involved in ionizing radiation-promoted liver carcinogenesis." Int J Clin Exp Med. 2010 Jul 31;3(3):211-22. PMID: 20827319 , Jul-2010